

Solution 1

Solution 1 (a)

Online Algorithm Implementation: Mean

```

1  """
2  Implementation of online algo from class
3  read data-1.txt using np.loadtxt(data-1.txt)
4  """
5  # load libraries
6  import numpy as np
7  import matplotlib.pyplot as plt
8  import seaborn as sns
9
10 # quick check data-1.txt has 5000 pts
11 # cat data-1.txt | wc -l -> 5000
12
13 # load data-1.txt
14 data = np.loadtxt('data-1.txt')
15
16 def online_algorithm_mean(data):
17     mean_i = []
18     dt = []
19     current_mean = 0
20     for t,x_t in enumerate(data):
21         dt.append(t+1)
22         current_mean = current_mean + ((x_t-current_mean)/(t+1))
23         mean_i.append(current_mean)
24     print(f"final mean: {mean_i[-1]}")
25     return dt, mean_i
26
27 def plot_mean_dt(dt,mean_i,true_mean):
28     fig,ax = plt.subplots(nrows=1,ncols=1,figsize=(8,10))
29     sns.lineplot(x=dt,y=mean_i,markers='o',ax=ax,color='blue',label='OA')
30     sns.lineplot(x=dt,y=[true_mean for i in
31         dt],markers='--',ax=ax,color='red',label='True Mean')
32     ax.set_xlabel('dt')
33     ax.set_ylabel('mean')
34     ax.set_ylim(min(mean_i)-1e5,true_mean+1e5)
35     print(min(mean_i))
36     title = f"Online Algorithm(OA) for Mean\n True Mean: {true_mean:.3f} \n OA Mean:
37         {mean_i[-1]:.3f}"
38     ax.set_title(title)
39     fig.savefig('hw3_q1.png')
40
41 # calc true mean and print it
42 true_mean = sum(data)/len(data)
43 print(true_mean)
44
45 dt, mean_i = online_algorithm_mean(data)
46 plot_mean_dt(dt,mean_i,true_mean)

```

Comments

The true mean is 2,991,834.673 and is identical to the final mean calculated via `online_algorithm_mean(data)` function.

Solution 1

Solution 1 (b)

Plot

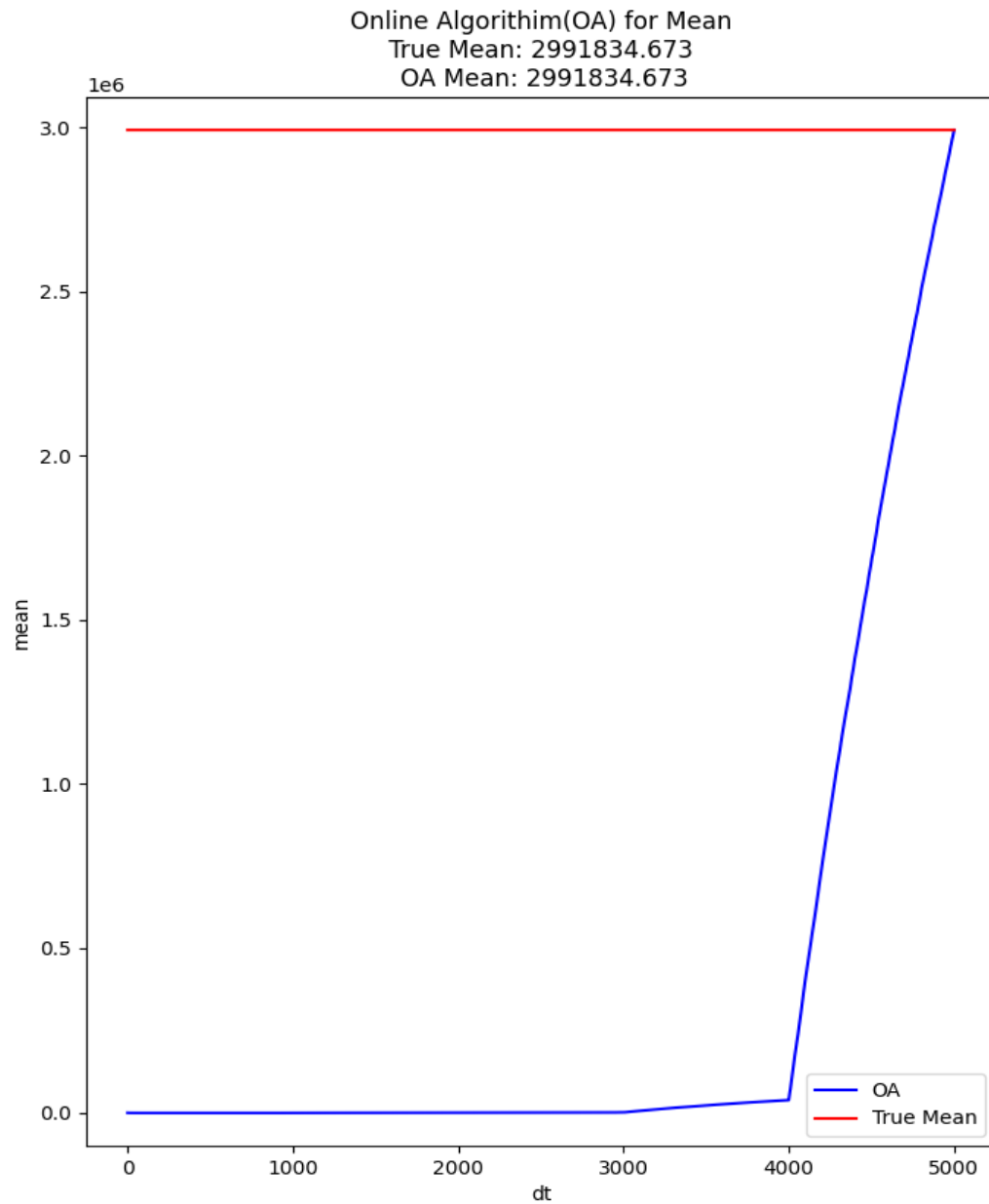


Figure 1: Figure shows all 5000 intermediate values of mean

Solution 2

Solution 2 (a)

pseudocode (ko zna??.. niko nista nej zna) for variance

remember $\text{Var}[X] = E[X^2] - E[X]^2$ ANSWER THIS: *how does online algo mean compare to true mean*

Solution 2

Solution 2 (b)**Online Algorithm Implementation: Variance**

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 """
4 Implementation of online algo from class
5 read data-1.txt using np.loadtxt(data-1.txt)
6 """
7 data = np.loadtxt
```

remember $\text{Var}[X] = E[X^2] - E[X]^2$ ANSWER THIS: *how does online algo variance compare to true variance*

Solution 3

Solution 3 (a)**True Median**

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 """
4 Implementation of online algo from class
5 read data-1.txt using np.loadtxt(data-1.txt)
6 """
7 data = np.loadtxt
```

ANSWER THIS: *report true median of data-1.txt*

Solution 3

Solution 3 (b)**Random Sample with Replacement Algorithm Implementation: Median**

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 """
4 Implementation of online algo from class
5 read data-1.txt using np.loadtxt(data-1.txt)
6 """
7 data = np.loadtxt
```

ANSWER THIS: *report true median of data-1.txt*

Solution 3

Solution 3 (c)**Random Sample with Replacement Algorithm Implementation: Median(sample: 500)**just repeat (b) with $n = 500$

Solution 4

something something something

Solution 5

Solution 5 (a)

something something something

Solution 5

Solution 5 (b)

something something something

Solution 6

Solution 6 (a)

something something something

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 """
4 code different
5 """
6 data = np.loadtxt
```

Solution 6

Solution 6 (b)

something something something

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 """
4 code different
5 """
6 data = np.loadtxt
```

Solution 7

Solution 7 (a)

something something something

Solution 7

Solution 7 (b)

something something something
